CONFIGURING VRF-LITE IN GNS3

Presented By
Cambell Xiao

Lab Purpose

This lab takes place in a simulated corporate environment where two networks are separated into different Virtual Routing and Forwarding zones. These two networks will be configured with the same networking scheme and general network topology, with the only difference being separate from the other through using the VRF. A host on each network should be able to contact only hosts on the same VRF.

Lab Background

Virtual Routing and Forwarding functions similar to a VLAN in a layer two switch. The main purpose of which is to keep traffic separate between multiple networks whilst running through the same router. Traffic running through one router can be separated depending on which VRF the traffic is coming from. This means that each VRF contains a separate routing table to maintain the isolation between these virtual networks.

The process of configuring VRF is not only extremely simple, but offers numerous benefits which includes security, performance, and network segmentation. Layer 3 routing can be incredibly beneficial for business not only because of the security benefits, but the simplicity in allowing similar networks to have the same subnet through one layer 3 device. The isolation between these VRFs means that all traffic originating from one VRF cannot access any other VRFs, which greatly simplifies the process of implementing security. Each VRF instance on a router contains its own routing table, which enhances the security of the network because traffic from one VRF can never access a separate VRF because a route doesn't exist to the other network. Not only that, but it's difficult to accidentally misconfigure VRF to accidentally overlap with another VRF instance's network because if a misconfiguration of that case were to happen, then typically no devices will have connectivity because VRF would not work due to the router having multiple interfaces on the same networking scheme. This would be incredibly desirable to businesses who want to enforce certain policies on a group of users, say the accounting group. All members of the accounting group can exist on one VRF subnet and a network policy can easily manage traffic only to that VRF without affecting any other ones.

To configure a VRF, each router needs to have a route to reach the destination networks. This means either running a routing protocol like OSPF or EIGRP or manually setting up static routes in each router. Without these routes, VRF cannot distinguish where to send traffic coming from one VRF and where to send traffic coming from another VRF. Routes are mandatory because if a router receives a packet, it will check its routing table and if no route to the destination exists, the router will forward the packet back towards the sender with a "destination host unreachable" message.

The concept of VRFs and VLANs may seem very similar in concept which may lead one to question the significance of a VRF when a VLAN can just be created instead. Both split up

traffic through the usage of their respective capabilities. VLANs divide traffic through the usage of tags whilst VRF divides traffic through virtual networks. The biggest difference between them is the layer in which they operate on; VLANs run on layer 2 and VRFs run on layer 3. Even though at a first glance, VLANS may seem superior due to their more granular ability to control and isolate traffic in the second layer, VRFs allow the division of virtual networks through IP addresses. This means that multiple networks connected to one router running VRF can have the same IP. Whilst VLANs are definitely more common than VRF, it is still an important option to consider. One of the main usages of VRF derives from internet service providers, who may need to isolate traffic between multiple customer's networks. A separate routing table for each customer would completely isolate them from each other, providing full security of information. In this case, dividing up networks into VLANs would not work because VLANs operate at layer 2 when dividing up networks requires layer 3 control.

There are two types of VRF. VRF-Lite and Full-VRF. VRF is usually used in conjunction with multi-protocol label switching, which is a method of routing using tags instead of the networks which many routing protocols use. VRF-Lite is the usage of VRF without any MPLS configurations. While Full-VRF is very highly scalable, VRF-Lite is not very scalable because for each VRF that exists on the network, a VRF command needs to be used on the router. The main benefit of VRF-Lite is the cost optimization of not having to add more routers to create more routing tables. This way, isolation between networks only needs to be done via one router's configurations instead of multiple.

Confirmation

VRF 1

Successful pings to devices in the same VRF

VM 1 to VM 3

```
Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::a882:b97e:4b84:95d0%4
   IPv4 Address. . . . . . . . . . . : 192.168.1.2
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.1.1

C:\Windows\system32>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:
Reply from 192.168.2.3: bytes=32 time=89ms TTL=124
Reply from 192.168.2.3: bytes=32 time=86ms TTL=124
Reply from 192.168.2.3: bytes=32 time=86ms TTL=124
Reply from 192.168.2.3: bytes=32 time=79ms TTL=124
```

VM 3 to VM 1

```
Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::520:cfea:9e75:552d%6
   IPv4 Address. . . . . . . . . . . : 192.168.2.2
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.2.1

C:\Users\mark>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time=79ms TTL=124
Reply from 192.168.1.3: bytes=32 time=80ms TTL=124
Reply from 192.168.1.3: bytes=32 time=85ms TTL=124
Reply from 192.168.1.3: bytes=32 time=82ms TTL=124
```

Failed pings to other VRFs

VM 1 to VM 4

```
Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::c8e0:3957:8135:128%4
   IPv4 Address. . . . . . . . . . . : 192.168.1.3
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.1.1

C:\Users\portland>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
```

VM 3 to VM 2

```
Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::622e:660:3b06:6bf4%5
   IPv4 Address. . . . . . . . . . . : 192.168.2.3
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.2.1

C:\Users\frank>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

VRF 2

<u>Successful pings to devices in the same VRF</u>

VM 2 to VM 4

```
Select C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.2965]
(c) Microsoft Corporation. All rights reserved.

C:\Users\portland>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::c8e0:3957:8135:128%4
   IPv4 Address. . . . . . . . . . . : 192.168.1.3
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.1.1

C:\Users\portland>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:
Reply from 192.168.2.2: bytes=32 time=81ms TTL=124
Reply from 192.168.2.2: bytes=32 time=79ms TTL=124
Reply from 192.168.2.2: bytes=32 time=83ms TTL=124
Reply from 192.168.2.2: bytes=32 time=86ms TTL=124

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 79ms, Maximum = 86ms, Average = 82ms

C:\Users\portland>
```

VM 4 to VM 2

```
Command Prompt
Microsoft Windows [Version 10.0.19045.2965]
(c) Microsoft Corporation. All rights reserved.

C:\Users\frank>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::622e:660:3b06:6bf4%5
   IPv4 Address. . . . . . . . . . . : 192.168.2.3
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.2.1

C:\Users\frank>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time=84ms TTL=124
Reply from 192.168.1.2: bytes=32 time=89ms TTL=124
Reply from 192.168.1.2: bytes=32 time=77ms TTL=124
Reply from 192.168.1.2: bytes=32 time=79ms TTL=124

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 77ms, Maximum = 89ms, Average = 82ms

C:\Users\frank>
```

<u>Failed pings to devices in other VRF</u>

VM 2 to VM 3

```
Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::c8e0:3957:8135:128%4
   IPv4 Address. . . . . . . . . . . : 192.168.1.3
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.1.1

C:\Users\portland>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
```

VM 4 to VM 1

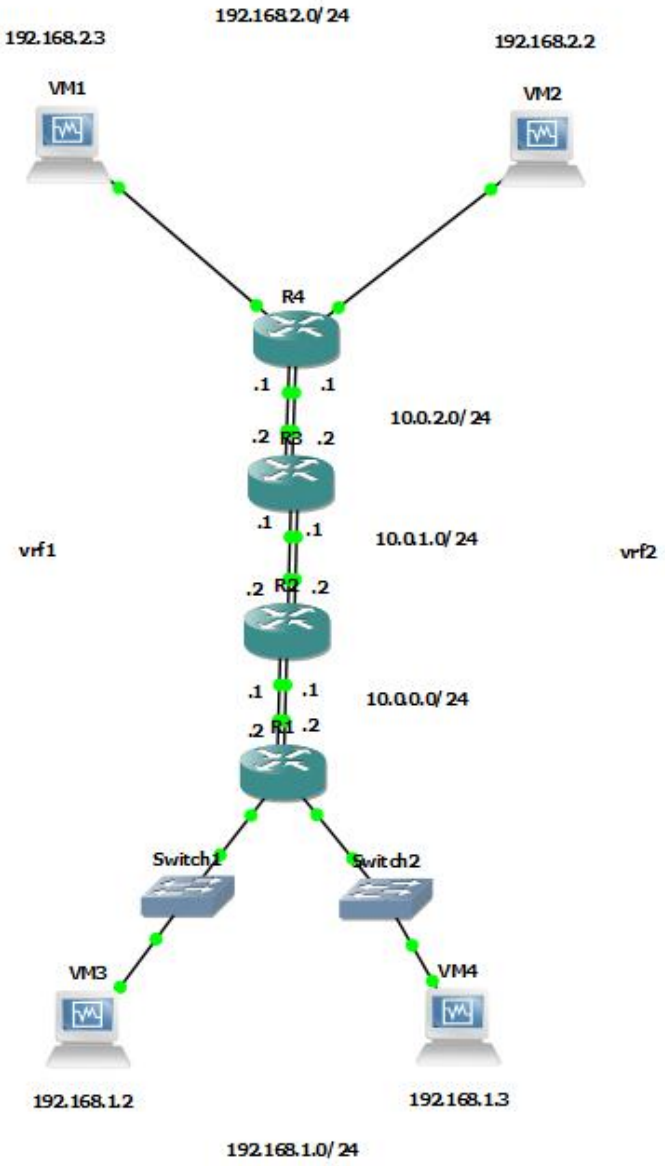
```
Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::622e:660:3b06:6bf4%5
   IPv4 Address. . . . . . . . . . . : 192.168.2.3
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.2.1

C:\Users\frank>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
```

Topology



Configurations

R1

```
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec

hostname R1
```

```
boot-start-marker
boot-end-marker

no aaa new-model
no ip icmp rate-limit unreachable
ip cef

ip vrf vrf1

ip vrf vrf2

no ip domain lookup
no ipv6 cef

multilink bundle-name authenticated

ip tcp synwait-time 5

interface FastEthernet0/0
no ip address
shutdown
duplex full

interface GigabitEthernet1/0
ip vrf forwarding vrf1
ip address 10.0.0.2 255.255.255.0
```

```
negotiation auto

interface GigabitEthernet2/0
ip vrf forwarding vrf2
ip address 192.168.1.1 255.255.255.0
negotiation auto

interface GigabitEthernet3/0
ip vrf forwarding vrf1
ip address 192.168.1.1 255.255.255.0
negotiation auto

interface GigabitEthernet4/0
ip vrf forwarding vrf2
ip address 10.0.0.2 255.255.255.0
negotiation auto

ip forward-protocol nd

no ip http server
no ip http secure-server
ip route vrf vrf1 0.0.0.0 0.0.0.0 GigabitEthernet1/0 10.0.0.1
ip route vrf vrf2 0.0.0.0 0.0.0.0 GigabitEthernet4/0 10.0.0.1
```

R2

```
version 15.2
service timestamps debug datetime msec
```

```
service timestamps log datetime msec

hostname R2

boot-start-marker
boot-end-marker
no aaa new-model
no ip icmp rate-limit unreachable
ip cef

ip vrf forward

ip vrf vrf1

ip vrf vrf2

no ip domain lookup
no ipv6 cef

multilink bundle-name authenticated

ip tcp synwait-time 5

interface FastEthernet0/0
no ip address
shutdown
duplex full
```

```
interface GigabitEthernet1/0
ip vrf forwarding vrf1
ip address 10.0.0.1 255.255.255.0
negotiation auto

interface GigabitEthernet2/0
ip vrf forwarding vrf2
ip address 10.0.1.2 255.255.255.0
negotiation auto

interface GigabitEthernet3/0
ip vrf forwarding vrf1
ip address 10.0.1.2 255.255.255.0
negotiation auto

interface GigabitEthernet4/0
ip vrf forwarding vrf2
ip address 10.0.0.1 255.255.255.0
negotiation auto

ip forward-protocol nd

no ip http server
no ip http secure-server
ip route vrf vrf1 0.0.0.0 0.0.0.0 GigabitEthernet3/0 10.0.1.1
```

```
ip route vrf vrf1 192.168.1.0 255.255.255.0 GigabitEthernet1/0
10.0.0.2
ip route vrf vrf2 0.0.0.0 0.0.0.0 GigabitEthernet4/0 10.0.0.2
ip route vrf vrf2 192.168.2.0 255.255.255.0 GigabitEthernet2/0
10.0.1.1
```

R3

```
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec

hostname R3

boot-start-marker
boot-end-marker

no aaa new-model
no ip icmp rate-limit unreachable
ip cef

ip vrf vrf1

ip vrf vrf2

no ip domain lookup
no ipv6 cef

multilink bundle-name authenticated
```

```
ip tcp synwait-time 5

interface FastEthernet0/0
no ip address
shutdown
duplex full

interface GigabitEthernet1/0
ip vrf forwarding vrf1
ip address 10.0.1.1 255.255.255.0
negotiation auto

interface GigabitEthernet2/0
ip vrf forwarding vrf2
ip address 10.0.2.2 255.255.255.0
negotiation auto

interface GigabitEthernet3/0
ip vrf forwarding vrf1
ip address 10.0.2.2 255.255.255.0
negotiation auto

interface GigabitEthernet4/0
ip vrf forwarding vrf2
ip address 10.0.1.1 255.255.255.0
negotiation auto
```

```
ip forward-protocol nd



no ip http server
no ip http secure-server
ip route vrf vrf1 0.0.0.0 0.0.0.0 GigabitEthernet3/0 10.0.2.1
ip route vrf vrf1 192.168.1.0 255.255.255.0 GigabitEthernet1/0
10.0.1.2
ip route vrf vrf2 0.0.0.0 0.0.0.0 GigabitEthernet4/0 10.0.1.2
ip route vrf vrf2 192.168.2.0 255.255.255.0 GigabitEthernet2/0
10.0.2.1
```

R4

```
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec

hostname R4

boot-start-marker
boot-end-marker

no aaa new-model
no ip icmp rate-limit unreachable
ip cef
```

```
ip vrf vrf1

ip vrf vrf2

no ip domain lookup
no ipv6 cef

multilink bundle-name authenticated

ip tcp synwait-time 5

interface FastEthernet0/0
no ip address
shutdown
duplex full

interface GigabitEthernet1/0
ip vrf forwarding vrf1
ip address 10.0.2.1 255.255.255.0
negotiation auto

interface GigabitEthernet2/0
ip vrf forwarding vrf2
ip address 192.168.2.1 255.255.255.0
negotiation auto

interface GigabitEthernet3/0
```

```
ip vrf forwarding vrf1
ip address 192.168.2.1 255.255.255.0
negotiation auto

interface GigabitEthernet4/0
ip vrf forwarding vrf2
ip address 10.0.2.1 255.255.255.0
negotiation auto

ip forward-protocol nd

no ip http server
no ip http secure-server
ip route vrf vrf1 0.0.0.0 0.0.0.0 GigabitEthernet1/0 10.0.2.2
ip route vrf vrf2 0.0.0.0 0.0.0.0 GigabitEthernet4/0 10.0.2.2
```

Lab Commands

ip vrf (vrf name) forwarding

Sets an interface to be designated to one VRF

ip vrf (vrf name)

creates a vrf

ip route vrf (vrf name) (destination IP) (subnet mask) (outgoing interface) (next-hop address)

Creates an IP route adding it to the routing table of the designated VRF.

Problems

Inconsistent VRF configurations on each router

VRFs are completely reliant on a a consistent router configuration on every router as this is the main purpose of VRF; to isolate networks. If one router is configured incorrectly then VRF does not work. We accidentally configured one router with a different VRF scheme compared to the others which resulted in our configurations not working.

Conclusion

While VRF may not be used consistently in everyday topologies, it is extremely useful in minor cases that do require it because the network would simply not work. The security and isolation of networks that VRF provides can sometimes outweigh the benefits of creating a VLAN, however the usefulness of VRF is greatly hindered by its lack of scalability. While it is helpful in small networks on the edge, it will not be helpful for core network functionalities.